Parallel & Perpendicular Gradients

Exam code: 9MA0

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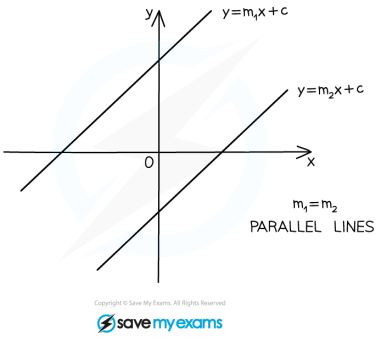
Parallel & perpendicular gradients

What are parallel lines?



- **Parallel** lines are equidistant meaning they never meet
- **Parallel** lines have **equal gradients**

Help me understand

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

$m_1 = m_2$

IF m IS SHOWN TO BE EQUAL IN TWO STRAIGHT LINE EQUATIONS THEY ARE PARALLEL

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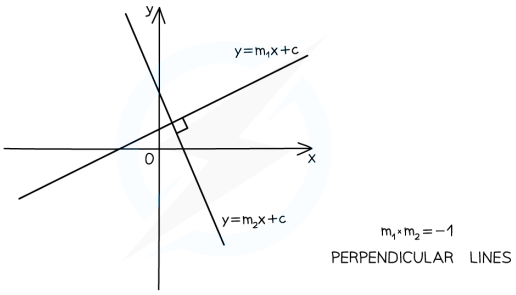
What are perpendicular lines?



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- **Perpendicular** lines meet at **right angles**
- The **product** of their **gradients** is -1

THE PRODUCT OF THE GRADIENTS OF PERPENDICULAR LINES IS −1

$m_1 \times m_2 = -1$

IF THE PRODUCT OF TWO GRADIENTS IS −1 THE GRAPHS ARE PERPENDICULAR

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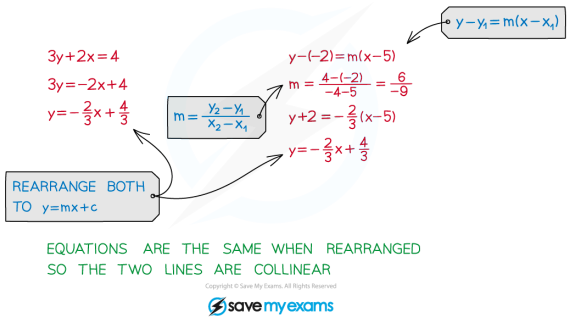
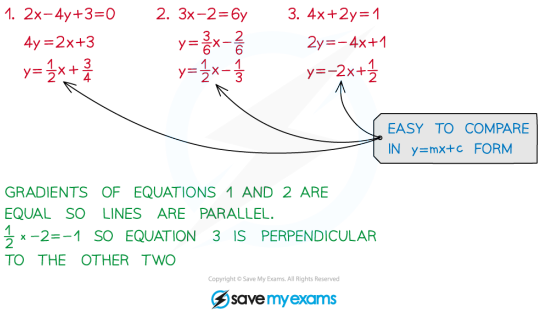
What does collinear mean?

- **Collinear** line segments are two line segments that are part of the **same** line
- They have **equal gradients**
  - They may **pass** through, or **meet** at, a common **point**
  - When they are extended, they become the same line
- **Rearrange** their equations into the same form and they will be identical



How do I tell if lines are parallel or perpendicular?

- Rearrange equations into the form **y = mx + c**
- **m** is the **gradient**



Examiner Tips and Tricks

- Exam questions are good at "hiding" parallel and perpendicular lines.
- For example a **tangent** and a **radius** are **perpendicular.**
- **Parallel lines** could be implied by phrases like "… at the same rate …"

Worked Example

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